electronics-toys_wave1_ORDERING

Wave 1 electronics toys: ordering guide (what to buy, where, in what order)

> Companion to MANUFACTURING_MANUAL.md. Main caveat for this project: AP1 board designed; MP3 module, display and buttons are bought-in. Prices and stock are from LCSC on the run date; vendors, shipping and customs rules change: verify before paying. General rules and vendor list: manuals/ORDERING_GUIDE.md.

1. Order of work

1. Read the manual and the caveat above.
2. Order the circuit boards and SMD assembly (section 2).
3. While the boards travel, order the other items (sections 3 and 5) and make the printed or laser-cut parts.
4. Assemble, program and test only after everything has arrived; follow the manual.

2. Circuit board

Board `ap1` has no release folder yet; see the manual for status.

3. Bought-in and mechanical items

See the parts and materials section of the manual.

4. Confidence (read before you order)

- **Order confidence: High**: all parts live-checked, none flagged.
- **Works-as-intended confidence: Low-Medium**: AP1 board added (ATmega328P, MP3 module socket, 4-digit display through two chained 74HC595s with a 7-LED ball lane, 7-button resistor ladder, lamp and motor drivers); all three apps now build for the board (about 6 kB each) and pass 76 host checks. The MP3 command codes are from memory and must be checked on the real module; the ladder thresholds are calculated, not measured.
- **First thing to do:** Bench-test one board per product: ladder readings, MP3 commands, display multiplexing.
- Rubric and the reasons for every project: `costing/CONFIDENCE.md`. These are judgements, not measurements: nothing here has been built.

5. Where to buy and whom to approach

Item type	India (fast, GST invoice)	Overseas (cheaper, 1 to 3 weeks, customs)
Printed circuit board and SMD assembly	JLCPCB is the practical choice for 5 to 100 boards; local PCB shops quote per project (ask for the Gerber ZIP price)	**JLCPCB** (jlcpcb.com), PCBWay (pcbway.com)

Item type	India (fast, GST invoice)	Overseas (cheaper, 1 to 3 weeks, customs)
Connectors, headers, passives, ICs by LCSC number	element14 India, Mouser India, DigiKey India (genuine, GST invoice, higher price); local electronics markets for common parts	**LCSC** (lcsc.com) with the same cart as JLCPCB
Modules: ESP32 dev boards, SIM7080G, LoRa (RA-08 / RAK3172), pH / EC / DO modules, relay modules, load cells, ultrasonic sensors	Robu.in, Robocraze, ThinkRobotics, Sunrom Electronics, Amazon.in	AliExpress, Seeed Studio, Waveshare
Motors, stepper drivers, wheels	Robu.in, Robocraze, local robotics shops	StepperOnline (omc-stepperonline.com)
Li-ion / LiFePO4 cells and packs, solar panels, MPPT controllers	Authorised battery and solar dealers, Amazon.in (buy protected cells with an invoice)	not recommended by post (shipping rules for batteries)
Laser-cut and 3D-printed parts	Local laser-cutting and 3D-printing shops (search IndiaMART, Justdial or maker spaces): send the SVG/DXF/STL from the repository and ask for a quote per piece	JLC3DP (jlc3dp.com), PCBWay 3D printing
Fasteners, tubing, hardware, enclosures	Local hardware, Robu.in, Amazon.in	n/a

How to approach a vendor: send the project name, the exact file (BOM, CPL, Gerber ZIP, STL or SVG), the quantity, and ask for the price, the lead time, the GST invoice and the warranty or return terms. Keep the reply for your records.